

Main Results: Magnetic Coupling in 7-AGNRs

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1 Introduction

This document summarizes the main results obtained for the calculation of the magnetic exchange coupling J in finite graphene nanoribbons (7-AGNRs) of different lengths, comparing the Tight-Binding model (effective Hubbard dimer) with the Mean-Field Hubbard (MFH) model.

2 Geometry

First, we verify the geometric construction of the system by removing atoms with coordination 1 at the zigzag edges to ensure a properly terminated finite molecule.

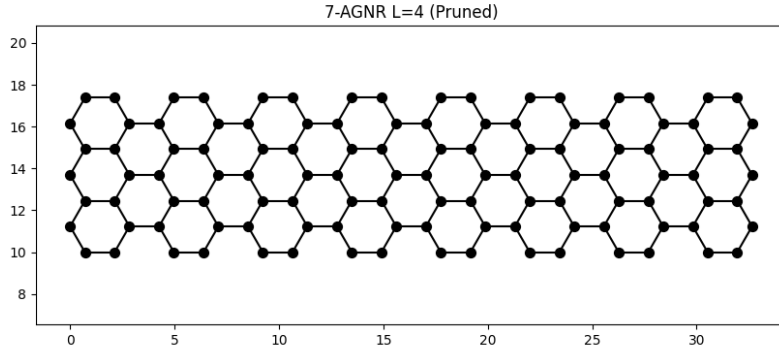


Figure 1: Geometry test for a short 7-AGNR ribbon ($L = 4$).

3 Spin Polarization (AFM and FM)

The ground state of the edges exhibits strong localization and spin polarization. Figure 2 shows the spin polarization profiles calculated using Mean-Field Hubbard (MFH) for the Antiferromagnetic (AFM, open-shell singlet) and Ferromagnetic (FM, triplet) configurations.

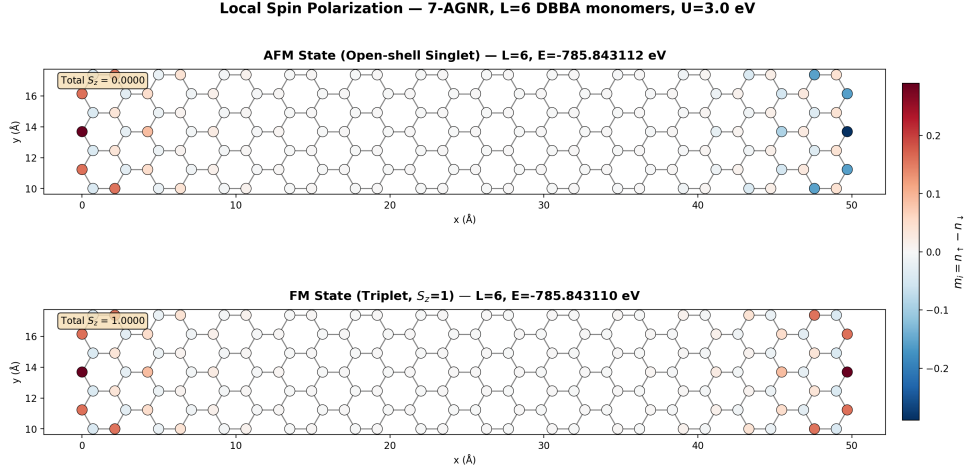


Figure 2: Spin distribution for the AFM and FM states calculated in a ribbon of length $L = 6$.

4 Magnetic Coupling J Comparison

We calculate the magnetic coupling J using two methods:

- The effective Hubbard dimer rule ($J = 4t_{\text{eff}}^2/U_{\text{eff}}$) derived from Tight-Binding.
- The energy difference between the FM and AFM states obtained from the MFH self-consistent cycle ($J = E_{\text{FM}} - E_{\text{AFM}}$).

The plot also includes experimental references, such as the superconducting gap 2Δ for Nb(110) and the typical thermal energy scale of an STM operating at 1.2 K.

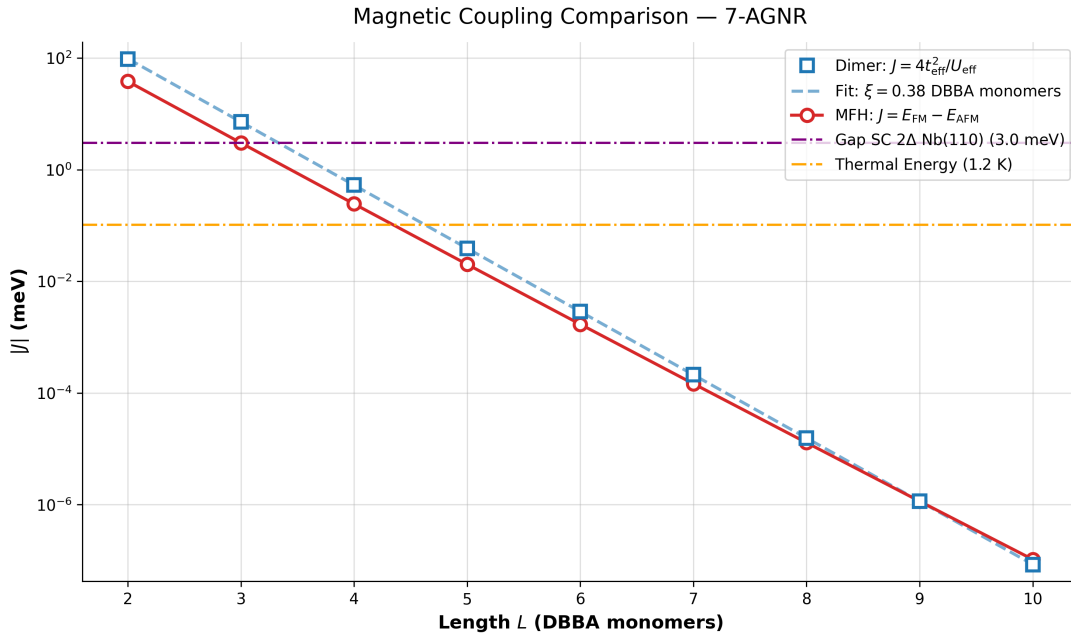


Figure 3: Comparison of the exchange coupling J extracted by both models as a function of the length L (number of DBBA monomers).

5 Energy Difference Between Models

To quantitatively observe the deviation between both methods, we analyze the absolute difference of the calculated couplings.

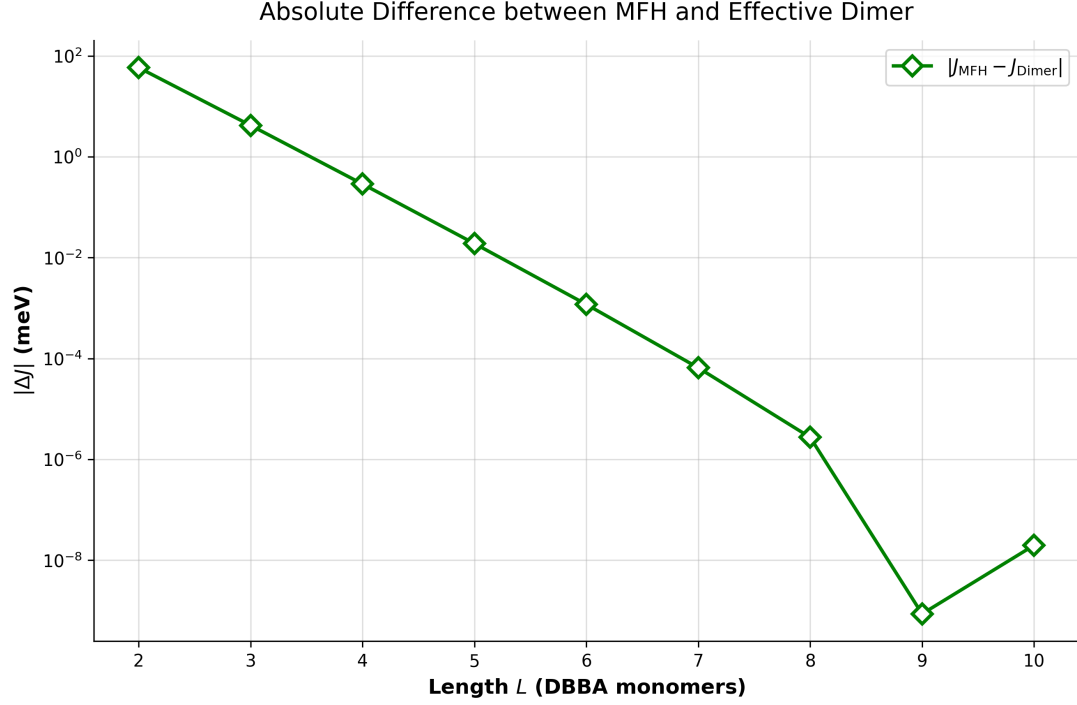


Figure 4: Absolute difference $|\Delta J| = |J_{\text{MFH}} - J_{\text{Dimer}}|$ as a function of length L .